CLI use

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- 1. Download the source code
- 2. Enable the optimal performance of the mainboard
  - 2.1. Enable the MAX power mode
  - 2.2. Enable the Jetson clocks
- 3. Predict with the CLI
  - 3.1. CLI usage syntax
  - 3.2. Image prediction
    - Effect preview
  - 3.3. Video prediction
    - Effect preview
  - 3.4. Real-time prediction
- References

1. Download the source code

```
git clone https://github.com/ultralytics/ultralytics.git
```

2. Enable the optimal performance of the mainboard

2.1. Enable the MAX power mode

Enabling the MAX Power Mode (the maximum power mode) on the Jetson ensures that all the CPU and the GPU cores are turned on:

```
sudo nvpmodel -m 0
```

2.2. Enable the Jetson clocks

Enabling the Jetson Clocks ensures that all the CPU and the GPU cores run at the maximum frequency:

```
sudo jetson_clocks
```

3. Predict with the CLI

CLI (Command-Line Interface, the command-line interface) is a way for a user to interact with a computer or a software program by typing text commands to run programs or perform tasks, instead of clicking icons or buttons in a graphical user interface (GUI).

3.1. CLI usage syntax

```
yolo TASK MODE ARGS

Where   TASK (optional) is one of [detect, segment, classify, pose, obb]
        MODE (required) is one of [train, val, predict, export, track, benchmark]
        ARGS (optional) are any number of custom 'arg=value' pairs like 'imgsz=320' that
override defaults.
```

3.2. Image prediction

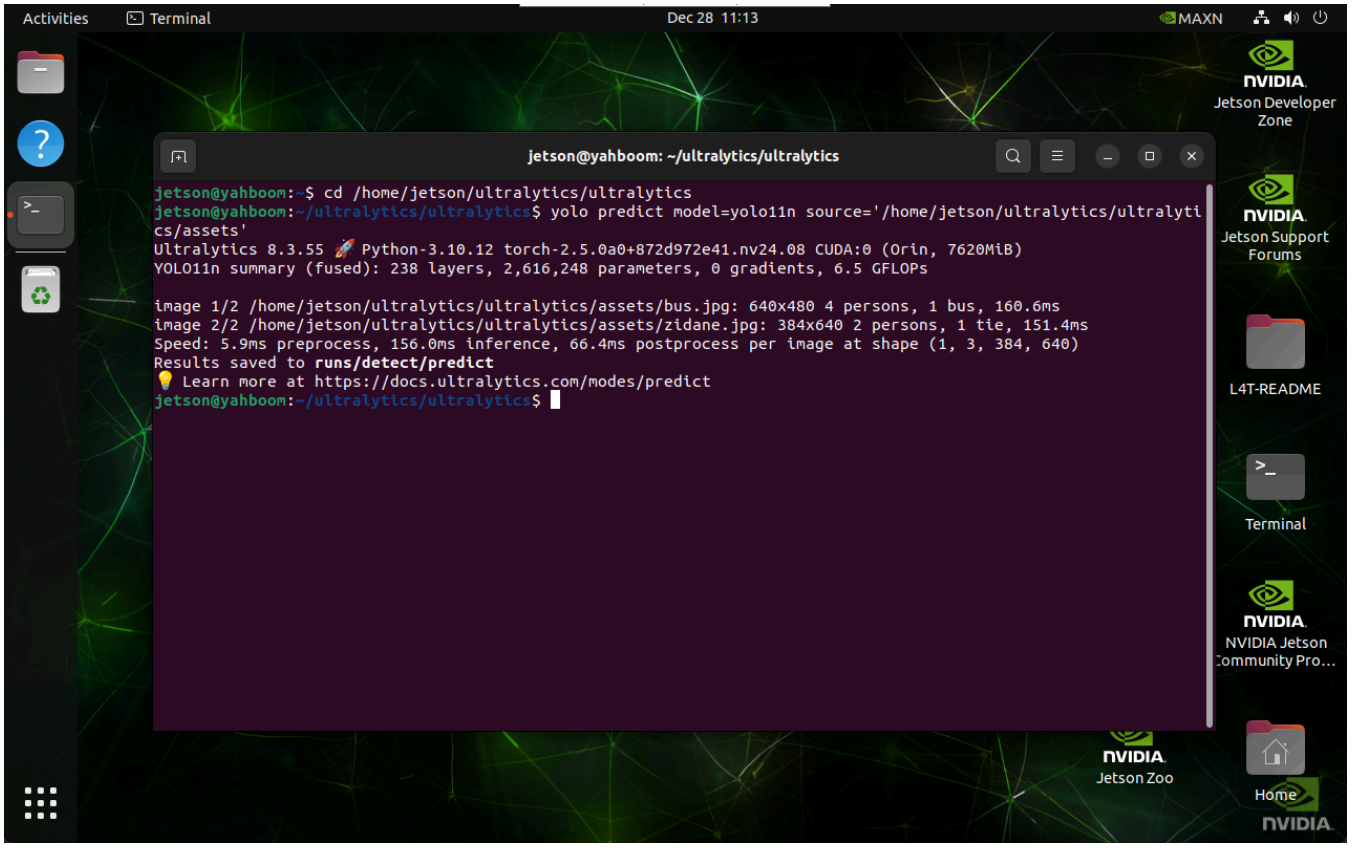
Use yolo11n.pt to predict the pictures that come with the ultralytics project: if the system does not find the corresponding model file in the directory in which the command is run, it downloads it automatically (if it cannot be downloaded, you can copy the model in yourself)

Enter the project folder:

```
cd /home/jetson/ultralytics/ultralytics/
```

Use yolo11n.pt to detect the pictures in the target folder and output the results:

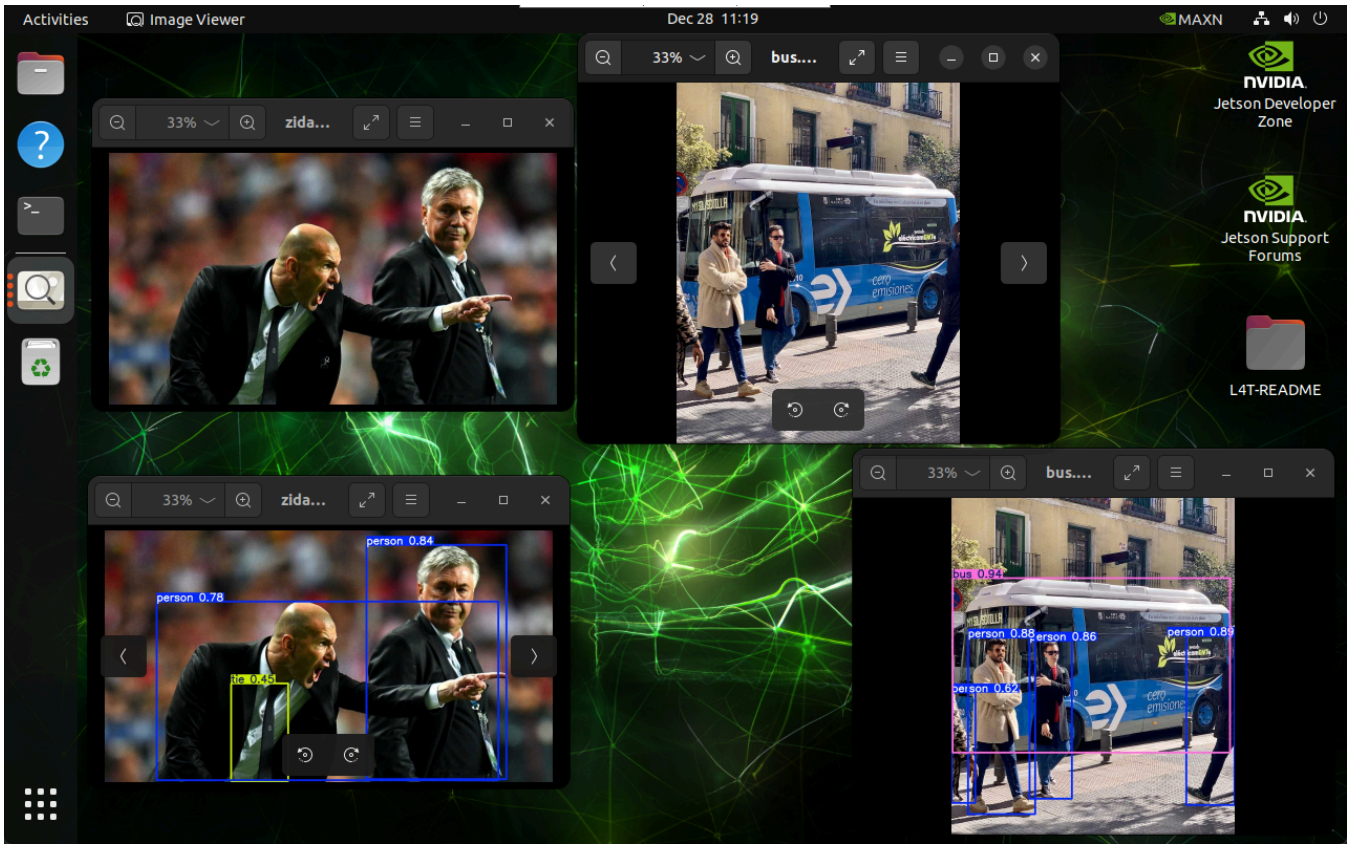
```
yolo predict model=yolo11n source='/home/jetson/ultralytics/ultralytics/assets'
```



Effect preview

The location of the videos that the yolo recognition outputs: /home/jetson/ultralytics/ultralytics/runs/detect;

Each time it runs, the predict under that folder is incremented automatically by 1



3.3. Video prediction

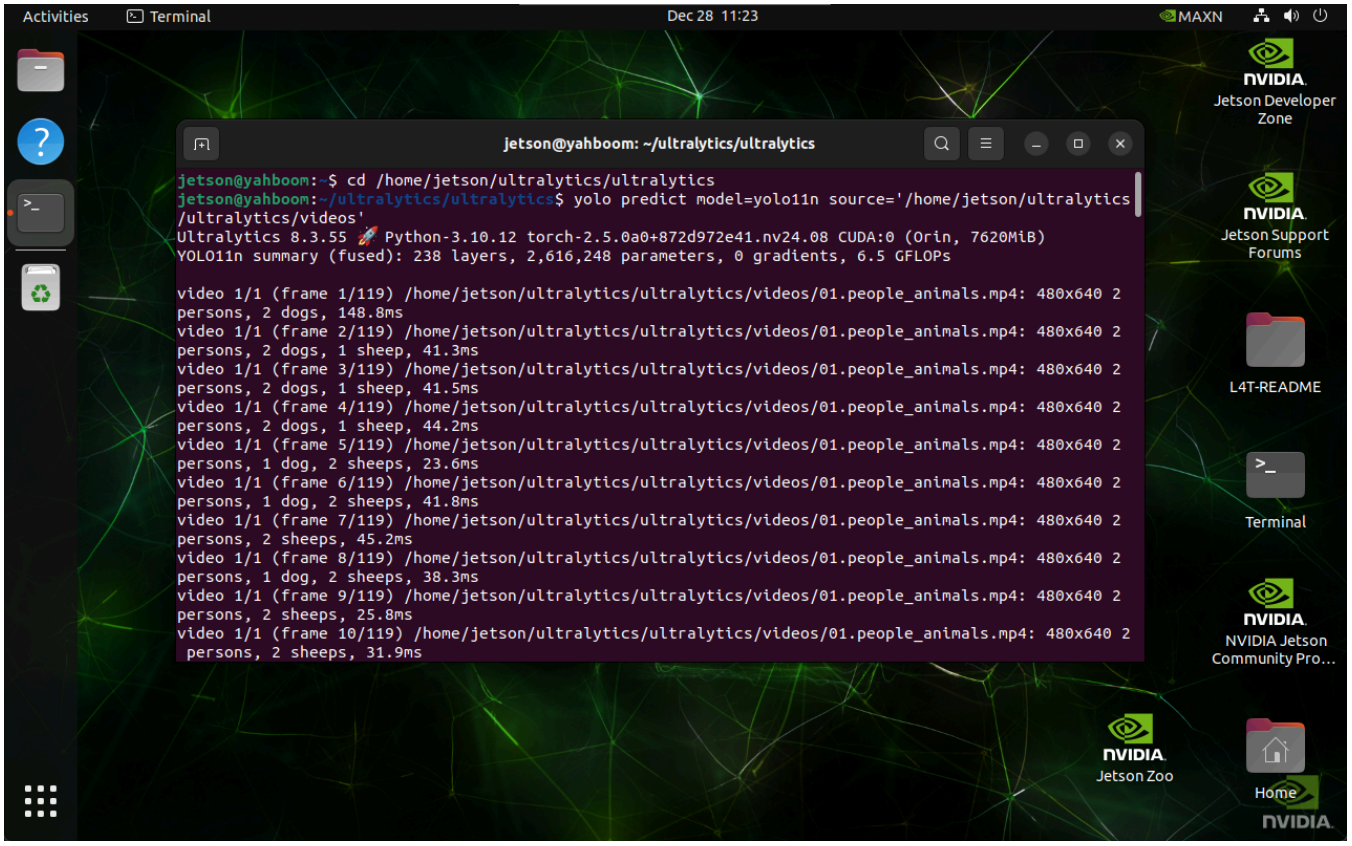
Use yolo11n.pt to predict the videos under the ultralytics project (not the videos that come with ultralytics): if the system does not find the corresponding model file in the directory in which the command is run, it downloads it automatically (if it cannot be downloaded, you can copy the model in yourself)

Enter the project folder:

```
cd /home/jetson/ultralytics/ultralytics
```

Use yolo11n.pt to detect the videos in the target folder and output the results:

```
yolo predict model=yolo11n source='/home/jetson/ultralytics/ultralytics/videos'
```



Effect preview

The location of the videos that the yolo recognition outputs: /home/jetson/ultralytics/ultralytics/runs/detect

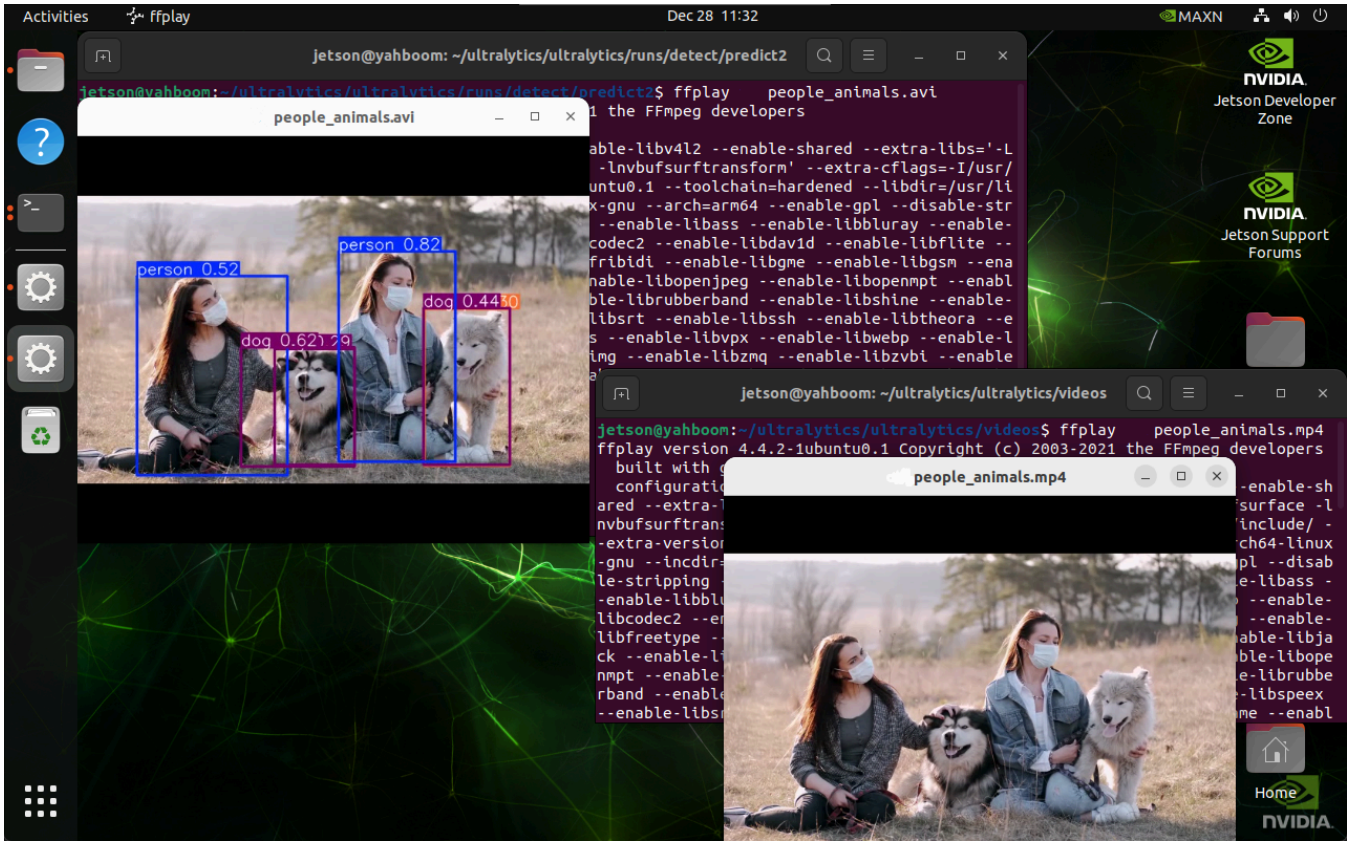
Each time it runs, the predict under that folder is incremented automatically by 1

You can watch the video with ffmpeg; install it with the following command:

```
sudo apt update
sudo apt install ffmpeg -y
```

Play the video:

```
ffplay <video_name>.avi
```



3.4. Real-time prediction

Use yolo11n.pt to predict the image of the USB camera: if the system does not find the corresponding model file in the directory in which the command is run, it downloads it automatically (if it cannot be downloaded, you can copy the model in yourself)

Enter the project folder:

```
cd /home/jetson/ultralytics/ultralytics
```

Use yolo11n.pt to detect the camera image and output the results: currently only the USB camera can use the CLI to predict the real-time image directly; for the CSI camera there is no information about using it directly as an input source.

Since the program has to load the model, the loading takes a while; please be patient.

```
yolo predict model=yolo11n.pt source=0 save=False show # Object detection
# Instance segmentation: yolo predict model=yolo11n-seg.pt source=0 save=False show
# Image classification: yolo predict model=yolo11n-cls.pt source=0 save=False show
# Pose estimation: yolo predict model=yolo11n-pose.pt source=0 save=False show
# Oriented object detection: yolo predict model=yolo11n-obb.pt source=0 save=False show
```

Click in the terminal and press the "Ctrl + C" shortcut to terminate the program!

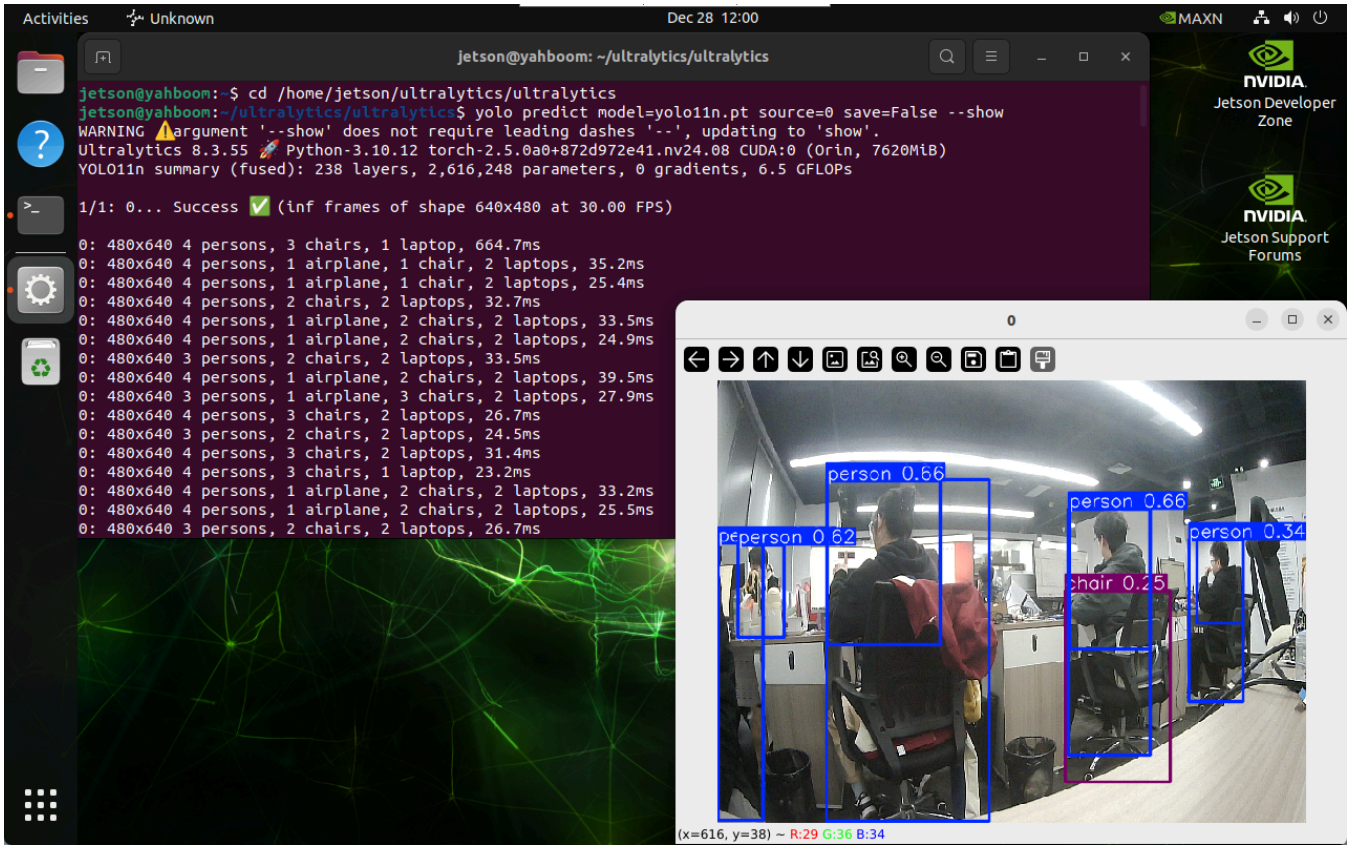
> **Parameter description**

`model`: specifies the YOLO model

`source`: specifies the recognition source: the number can be switched with several cameras

`save=False`: disables saving the results

`show`: displays in real time



References

https://github.com/ultralytics/ultralytics

https://docs.ultralytics.com/usage/cli/